

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE - MEA & MEPS-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E5

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Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2024

SECTION-A (PHYSICS)

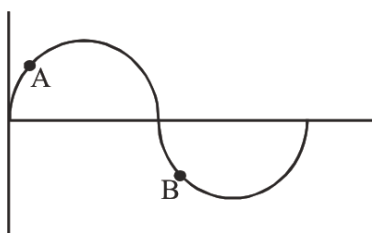
1. The frequency of man's voice is 300 Hz and its wavelength is 1 meter. If the wavelength of a child's voice is 1.5m, then the frequency of the child's voice is :-

- (1) 200 Hz
- (2) 150 Hz
- (3) 400 Hz
- (4) 450 Hz

2. The equation of a travelling wave is $y = 60 \cos(1800t - 6x)$ where y is in microns, t in seconds and x in metres. The ratio of maximum particle velocity to velocity of wave propagation is :-

- (1) 3.6×10^{-11}
- (2) 3.6×10^{-6}
- (3) 3.6×10^{-4}
- (4) 3.6

3. Consider the snapshot of a wave traveling in positive x -direction



- (1) The particle A is moving in $-ve$ y -direction and particle B is moving in $+ve$ y -direction
- (2) The particle B is moving in $-ve$ y -direction and particle A is moving in $+ve$ y -direction
- (3) Both are moving in the $+ve$ y -direction
- (4) Both are moving in the $-ve$ y -direction

4. The equation of a wave on a string of linear density 0.04 kg m^{-1} is given by.

$$y = 0.02(\text{m}) \sin \left[2\pi \left(\frac{t}{0.04(\text{s})} - \frac{x}{0.50(\text{m})} \right) \right]$$

The tension in the string is.

- (1) 6.25 N
- (2) 4.0 N
- (3) 12.5 N
- (4) 0.5N

5. Two waves represented by the following equations are travelling in the same medium

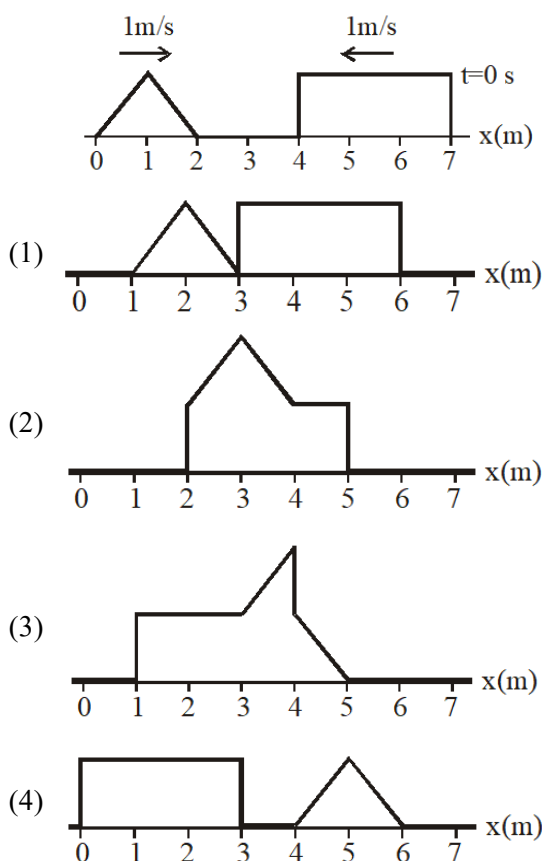
$$y_1 = 5 \sin 2\pi(75t - 0.25x)$$

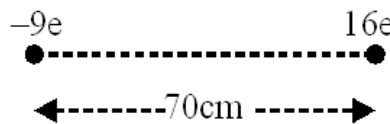
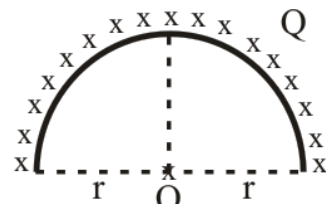
$$y_2 = 10 \sin 2\pi(150t - 0.50x)$$

The intensity ratio $\frac{I_1}{I_2}$ of the two wave is

- (1) 1 : 2
- (2) 1 : 4
- (3) 1 : 8
- (4) 1 : 16

6. Two pulses on a string approach each other at speeds of 1 m/s. What is the shape of the string at $t = 3 \text{ s}$:-



7. Two waves with same amplitude and wavelength travel in opposite directions with equal speed in a medium. When the waves are superimposed the phenomenon observed is
- Stationary waves
 - Harmonic travelling waves
 - Beats
 - Polarisation
8. When the temperature of an ideal gas is increased by 600 K, the velocity of sound in the gas become $\sqrt{3}$ times the initial velocity in it. The initial temperature of the gas is :-
- -73°C
 - 27°C
 - 127°C
 - 327°C
9. Intensity level 20 cm from a point source of sound is 80 dB. If there is no loss of acoustic power in air and intensity of threshold hearing is 10^{-12} Wm^{-2} then what is the intensity level at a distance of 400 cm from source :-
- zero
 - 54 dB
 - 64 dB
 - 44 dB
10. The circular scale of a micrometer has 200 divisions and pitch of 2mm. Find the measured value of thickness of a thin sheet.
- 
- 3.41 mm
 - 6.41 mm
 - 3.46 mm
 - 3.51 mm
11. The potential energy of a particle varies with distance x from a fixed point as $U = \frac{A\sqrt{x}}{x + B^2}$. Where A and B are constants. Then, the dimension of AB^2 :
- $[ML^{3/2}T^{-2}]$
 - $[M^1L^{7/2}T^{-2}]$
 - $[M^2L^{5/2}T^{-2}]$
 - $[M^0L^0T^0]$
12. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are:-
- 4, 4, 2
 - 5, 1, 2
 - 5, 1, 5
 - 5, 5, 2
13. If unit of time is doubled then how many times the unit of length must become so that acceleration due to gravity does not change ?
- 4
 - 2
 - 8
 - $\sqrt{2}$
14. body has -80 microcoulomb of charge. Number of additional electrons on it will be :
- 8×10^{-5}
 - 80×10^{15}
 - 5×10^{14}
 - 1.28×10^{-17}
15. An infinite number of charges, each of charge $1 \mu\text{C}$, are placed on the x-axis with co-ordinates $x = 1, 2, 4, 8, \dots, \infty$. If a charge of 1 C is kept at the origin, then what is the net force acting on 1 C charge :
- 9000 N
 - 12000 N
 - 24000 N
 - 36000 N
16. Two charges are placed as shown in figure. Where should a third charge be placed so that it remains at rest.
- 
- 280 cm from 16e
 - 40 cm from 16e
 - 30 cm from 9e
 - Both (1) and (2)
17. Total charge Q is uniformly distributed over rim of a semicircular arc of radius r . Electric field at centre O is
- 
- $\frac{KQ}{r}$
 - $\frac{2KQ}{\pi r^2}$
 - $\frac{2KQ}{r}$
 - $\frac{Q}{2\pi \epsilon_0 r}$

18. Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio $1 : 4$, the current density in the two wires will be in the ratio :

(1) $1 : 2$ (2) $2 : 1$ (3) $4 : 1$ (4) $1 : 1$

19. Charge passing through a conductor of cross-section area $A = 0.3 \text{ m}^2$ is given by $q = 3t^2 + 5t + 2$ in coulomb, where t is in second. What is the value of drift velocity at $t = 2\text{s}$?

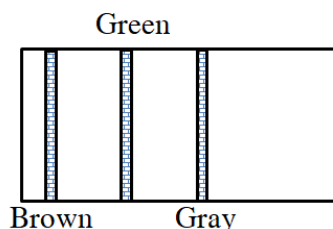
(Take, $n = 2 \times 10^{25} / \text{m}^3$)

(1) $0.77 \times 10^{-5} \text{ m/s}$ (2) $1.77 \times 10^{-5} \text{ m/s}$
(3) $2.08 \times 10^5 \text{ m/s}$ (4) $0.57 \times 10^5 \text{ m/s}$

20. The specific resistance of the material of a wire is ρ and its volume is 3m^3 and its resistance is 3Ω . The length of the wire will be :-

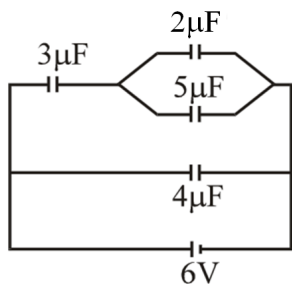
(1) $\sqrt{\frac{1}{\rho}}$ (2) $\frac{3}{\sqrt{\rho}}$ (3) $\frac{\sqrt{3}}{\rho}$ (4) $\frac{\rho}{\sqrt{3}}$

21. Value of resistance shown in the figure is



- (1) 1500 mega ohms $\pm 20\%$
(2) $1500 \pm 10\%$
(3) 15000 mega ohms
(4) 15 mega ohms

22. In the circuit shown in figure, the ratio of charges on $5\mu\text{F}$ and $4\mu\text{F}$ capacitor is :-



- (1) $4/5$ (2) $3/5$ (3) $3/8$ (4) $1/2$

23. The energy stored in a parallel plate condenser of plate separation d and plate area of cross-section A such that the uniform electric field between the plates is E , :-

(1) $\epsilon_0 E^2 A d$ (2) $\frac{1}{2} \epsilon_0 E^2 A d$
(3) $\frac{1}{2} \epsilon_0 E^2 / A d$ (4) $\epsilon_0 E^2 / A d$

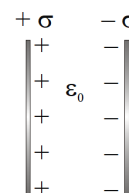
24. Potential of a drop is 1600 volt. If this drop is split into 64 drops then calculate the potential of smaller drop :

(1) 400 V (2) 200 V (3) 100 V (4) 800 V

25. A capacitor of capacitance $1\mu\text{F}$ and potential 10V is connected to a battery of emf 12V, then energy supplied by the battery after connection :

(1) $24 \mu\text{J}$ (2) $20 \mu\text{J}$ (3) $10 \mu\text{J}$ (4) $12 \mu\text{J}$

26. Two large metal are placed parallel to each other. The inner surfaces of plates are charged by $+\sigma$ and $-\sigma$ (Coulomb/ m^2). The outer surfaces are neutral. The electric field is in the region between the plates and outside the plates :-



(1) $\frac{2\sigma}{\epsilon_0}$, $\frac{\sigma}{\epsilon_0}$ (2) $\frac{2\sigma}{\epsilon_0}$, zero
(3) $\frac{\sigma}{\epsilon_0}$, zero (4) zero, $\frac{2\sigma}{\epsilon_0}$

27. The energy density in a parallel plate capacitor is given as $2.1 \times 10^{-9} \text{ J/m}^3$. The value of the electric field in the region between the plates is :

(1) 2.1 NC^{-1} (2) 21.6 NC^{-1}
(3) 72 NC^{-1} (4) 8.4 NC^{-1}

28. The force between the plates of a parallel plate capacitor of capacitance C and distance of separation of the plates d with a potential difference V between the plates, is :-

(1) $\frac{CV^2}{2d}$ (2) $\frac{C^2V^2}{2d^2}$ (3) $\frac{C^2V^2}{d^2}$ (4) $\frac{V^2d}{C}$

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29. A charged parallel plate capacitor of distance (d) has U_0 energy. A slab of dielectric constant (K) and thickness (d) is then introduced between the plates of the capacitor. The new energy of the system is given by :

- (1) KU_0
- (2) K^2U_0
- (3) $\frac{U_0}{K}$
- (4) $\frac{U_0}{K^2}$

30. Two condenser of capacity $0.3 \mu\text{F}$ and $0.6 \mu\text{F}$ respectively are connected in series. The combination is connected across a potential of 6 volts. The ratio of energies stored by the condensers will be :-

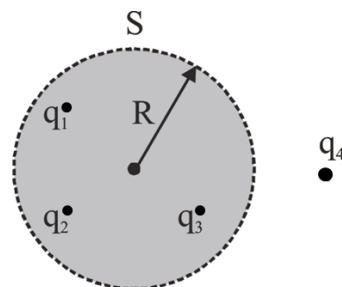
- (1) $1/3$
- (2) 2
- (3) $1/4$
- (4) 4

31. A parallel plate capacitor is charged by a battery. After charging the capacitor, battery is disconnected and distance between the plates is decreased then which of the following statement is correct ?

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- (1) electric field does not remain constant
- (2) potential difference is increased
- (3) the capacitance decreases
- (4) the stored energy decreases

32. q_1, q_2, q_3 and q_4 are point charges located at points as shown in the figure and S is a spherical Gaussian surface of radius R . Which of the following is true according to the Gauss's law

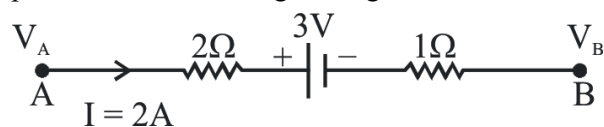


- (1) $\oint_s (\vec{E}_1 + \vec{E}_2 + \vec{E}_3) \cdot d\vec{A} = \frac{q_1 + q_2 + q_3}{2\epsilon_0}$
- (2) $\oint_s (\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4) \cdot d\vec{A} = \frac{q_1 + q_2 + q_3}{\epsilon_0}$
- (3) $\oint_s (\vec{E}_1 + \vec{E}_2 + \vec{E}_3) \cdot d\vec{A} = \frac{q_1 + q_2 + q_3 + q_4}{2\epsilon_0}$
- (4) None of the above

33. At following temperature, resistance of a copper wire will be triple of its resistance at 0°C ($\alpha = \frac{1}{273} ^\circ\text{C}^{-1}$)

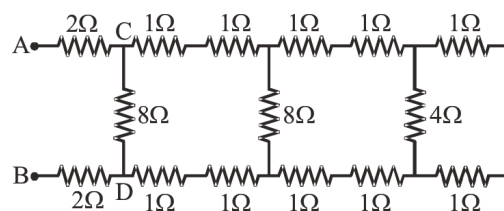
- (1) 273 K
- (2) 546 K
- (3) 819 K
- (4) 1092 K

34. The potential difference ($V_A - V_B$) between the points A and B in the given figure is :



- (1) $+9\text{V}$
- (2) -3V
- (3) $+3\text{V}$
- (4) $+6\text{V}$

35. In the figure shown, the total resistance between A and B is :



- (1) 12Ω
- (2) 4Ω
- (3) 6Ω
- (4) 8Ω

SECTION-B (PHYSICS)

36. Two waves having equation

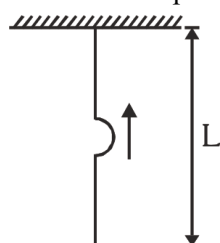
$$x_1 = a \sin(\omega t + f_1)$$

$$x_2 = a \sin(\omega t + f_2)$$

If in the resultant wave the frequency and amplitude remains equals to amplitude of superimposing waves. Then phase diff. between them :-

- (1) $\frac{\pi}{6}$ (2) $\frac{2\pi}{3}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{3}$

37. A thick uniform rope of length L is hanging from a rigid support. A transverse wave of wavelength λ_0 is set up in the middle of the rope. The wavelength of the wave as it reaches the topmost point is :-

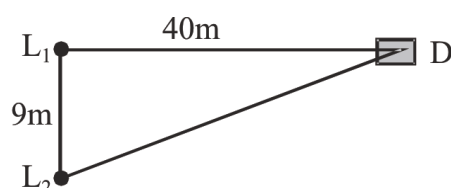


- (1) $2\lambda_0$ (2) $\sqrt{2}\lambda_0$ (3) $\lambda_0/\sqrt{2}$ (4) λ_0

38. Waves from two sources superimpose on each other at a particular point. Amplitude and frequency of both the waves are equal. The ratio of intensities when both waves reach in the same phase and they reach with the phase difference of 90° will be

- (1) 1 : 1 (2) $\sqrt{2} : 1$ (3) 2 : 1 (4) 4 : 1

39. Two loudspeakers L_1 and L_2 , driven by a common oscillator and amplifier, are arranged as shown in fig. The frequency of the oscillator is gradually increased from zero and the detector at D records a series of maxima and minima. If speed of sound is 330 m/s, then the frequency at which the first maximum is observed is :-

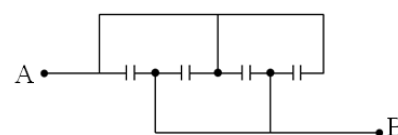


- (1) 165 Hz (2) 330 Hz
(3) 495 Hz (4) 660 Hz

40. If P represents radiation pressure, c represents speed of light and Q represents radiation energy striking a unit area per second, then non-zero integers x , y and z such that $P^x Q^y c^z$ is dimensionless, can be :-

- (1) $x = 1, y = 1, z = -1$
(2) $x = 1, y = -1, z = 1$
(3) $x = -1, y = 1, z = 1$
(4) $x = 1, y = 1, z = 1$

41. Four condensers are joined as shown in the figure below. The capacity of each is $8 \mu\text{F}$. The equivalent capacity between the points A and B will be



- (1) $32 \mu\text{F}$ (2) $2 \mu\text{F}$
(3) $8 \mu\text{F}$ (4) $16 \mu\text{F}$

42. When a dielectric of susceptibility 9 is placed in an electric field then electrical polarization of 10 nC/m^2 is induced. When another dielectric is placed in same electric field then electrical polarization of 20 nC/m^2 is induced then dielectric constant of the second dielectric is :-

- (1) 18 (2) 19
(3) 17 (4) 21

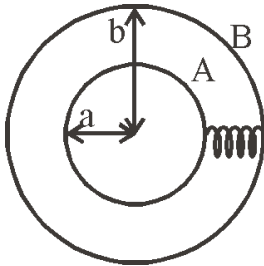
43. A capacitor of capacitance C is filled with a dielectric of constant K . It is connected to a battery of emf V then work done in removing the dielectric is : TG: @Chalnaayaaar

- (1) $\frac{1}{2} CV^2 \left(\frac{1}{K^2} - K \right)$
(2) $\frac{1}{2} CV^2 (1 - K)$
(3) $\frac{1}{2} CV^2 (K^2 - 1)$
(4) $\frac{1}{2} CV^2 \left(1 - \frac{1}{K^2} \right)$

44. An uncharged capacitor having capacitance C is connected across a battery of emf V . Now the capacitor is disconnected and then reconnected across the same battery but with reversed polarity. Then which of the statements is incorrect ?

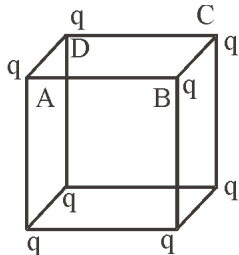
- (1) After reconnecting, charge delivered by the battery is $2CV$.
- (2) After reconnecting, no energy is supplied by battery.
- (3) After reconnecting, whole of the energy supplied by the battery is converted into heat.
- (4) After reconnecting, thermal energy produced in the circuit will be equal to $2CV^2$.

45. Two spherical conductors A and B of radii a and b ($b > a$) are placed concentrically in air. The two are connected by a copper wire as shown in figure. Then the equivalent capacitance of the system is :-



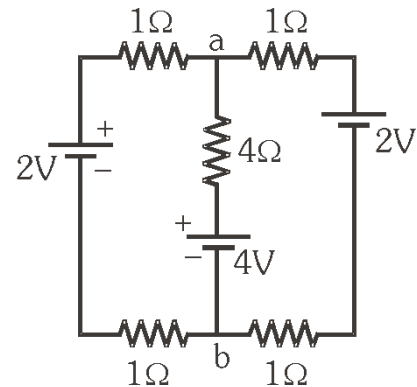
- (1) $4\pi\epsilon_0 \frac{ab}{b-a}$
- (2) $4\pi\epsilon_0(a+b)$
- (3) $4\pi\epsilon_0 b$
- (4) $4\pi\epsilon_0 a$

46. Eight point charges having magnitude q are fixed at vertices of a cube. The electric flux through square surface ABCD of the cube is :-



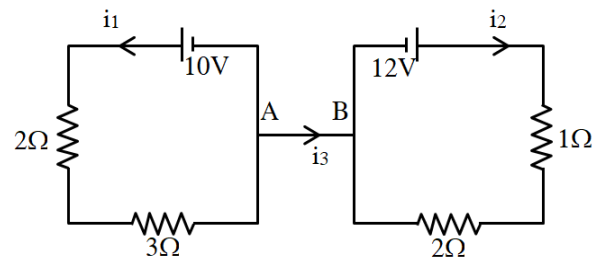
- (1) $\frac{q}{24\epsilon_0}$
- (2) $\frac{q}{12\epsilon_0}$
- (3) $\frac{q}{6\epsilon_0}$
- (4) $\frac{q}{3\epsilon_0}$

47. For the circuit shown in figure, the potential difference between points a and b is :-



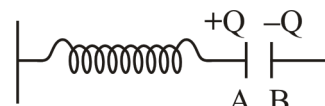
- (1) 2.4 V
- (2) 2.8 V
- (3) 2 V
- (4) 3.2 V

48. In the given circuit



- (1) $i_3 = 0$
- (2) $i_2 > 4A$
- (3) $i_1 < 2A$
- (4) $i_1 + i_2 = i_3$

49. Consider the situation as shown in the figure. Plate B is fixed and force constant of spring is K . If charge $+Q$ is placed on plate A & charge $-Q$ on plate B. Find extension in the spring. A is area of plate of capacitor. This system is placed on smooth floor.



- (1) $\frac{Q^2}{A\epsilon_0 K}$
- (2) $\frac{2Q^2}{A\epsilon_0 K}$
- (3) $\frac{Q^2}{2A\epsilon_0 K}$
- (4) $\frac{3Q^2}{A\epsilon_0 K}$

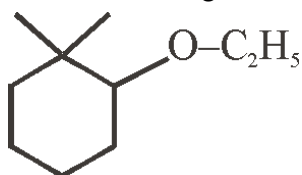
50. A spherical capacitor of inner radii r_1 and outer radii r_2 has capacitance C_1 . A spherical conductor of radius r_2 has a capacitance C_2 .

- (1) C_1 may equal to C_2
- (2) C_1 always greater than C_2
- (3) C_1 always lesser than C_2
- (4) None of the above

Topic : p-BLOCK (UPTO OXYGEN FAMILY), CLASSIFICATION AND NOMENCLATURE, ISOMERISM (TILL TAUGHT) [BACK UNIT] : PERIODIC TABLE TG: @Chalnaayaaar

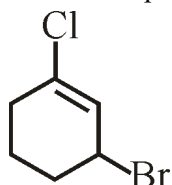
SECTION-A (CHEMISTRY)

51. What is the IUPAC name of given compound is:-



- (1) 1-ethoxy-2, 2-dimethyl cyclohexane
- (2) 2-ethoxy-1, 1-dimethyl cyclohexane
- (3) 2, 2-dimethyl cyclohexane-1-ethoxy
- (4) 1,1-dimethyl cyclohexane-2-ethoxy

52. The IUPAC name of the compound is :

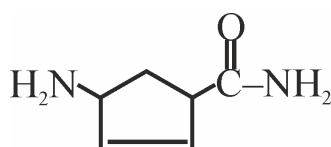


- (1) 6-bromo-2-chlorocyclohexene
- (2) 3-bromo-1-chlorocyclohexene
- (3) 1-bromo-3-chlorocyclohexene
- (4) 2-bromo-6-chlorocyclohex-1-ene

53. A pair of homologues is :

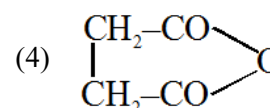
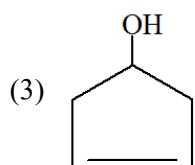
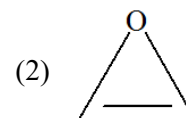
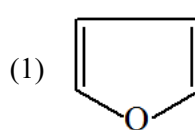
- (1) $\text{CH}_3\text{--CH=CH--CH}_3$ & $\text{CH}_3\text{--CH}_2\text{--CH=CH}_2$
- (2) $\text{CH}_3\text{--CH=O}$ and $\text{CH}_3\text{--O--CH}_3$
- (3) $\text{CH}_3\text{--CH}_2\text{--O--CH}_3$ and $\text{CH}_3\text{--O--CH}_3$
- (4) $\text{CH}_3\text{--CH}_2\text{--NH}_2$ and $\text{CH}_3\text{--CH}_2\text{--NH--CH}_3$

54. IUPAC name for :

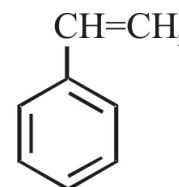


- (1) 4-Amino cyclopent-2-enamide
- (2) 3-Amino cyclopent-4-enamide
- (3) 4-Amino cyclopent-2-enecarboxamide
- (4) 3-Amino cyclopent-4-enecarboxamide

55. The Heterocyclic compound which have one olefinic bond and one hetero atom :-

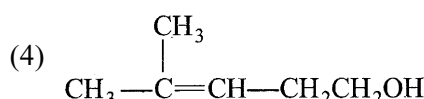
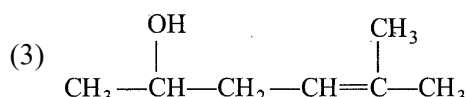
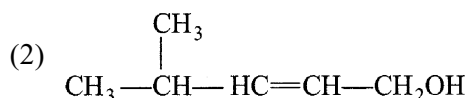


56. The number of 1° , 2° , 3° H, respectively in the given compound are :-



- (1) 2, 6, 1 (2) 2, 6, 0 (3) 2, 7, 1 (4) 2, 5, 0

57. The structure of 4-methylpent-2-en-1-ol is:



58. Match the column

	Column-I (Common Name)		Column-II (Structural Formula)
(A)	Acetone	(p)	$\text{CH}_3\text{--CHO}$
(B)	Acetaldehyde	(q)	$\text{CH}_3\text{--C(=O)--CH}_3$
(C)	Crotonaldehyde	(r)	$\text{CH}_2\text{=CH--CHO}$
(D)	Acrylaldehyde	(s)	$\text{CH}_3\text{--CH=CH--CHO}$

(1) A-q; B-p; C-r; D-s (2) A-q; B-p; C-s; D-r

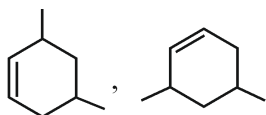
(3) A-p; B-s; C-r; D-q (4) A-q; B-r; C-s; D-p

59. (I) $\text{CH}_3 - \text{CH}_2 - \text{NH} - \text{CHO}$;
 (II) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CHO} \\ | \\ \text{NH}_2 \end{array}$

Which type of isomerism is observed between I and II.

- (1) Chain isomers (2) Position isomers
 (3) Functional isomers (4) Metamers

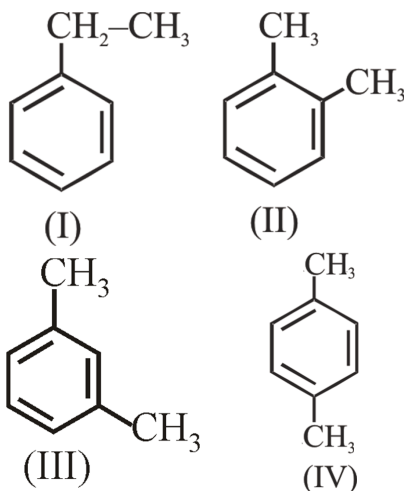
60.



Relation between above compounds is:

- (1) Position isomers (2) Chain isomers
 (3) Identical (4) Functional isomers

61. C_8H_{10} has 4 benzenoid structure :



- (1) I and II are position isomers
 (2) II and III are chain isomers
 (3) I and III are chain isomers
 (4) I and IV are position isomers

62. Which of the following pair does not related as position isomer ?

- (1) $\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3 - \text{CH} - \text{Cl} \end{array}$ and $\begin{array}{c} \text{OH} \\ | \\ \text{CH}_2 - \text{CH}_2 - \text{Cl} \end{array}$
 (2) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$ and $\begin{array}{c} \text{NH}_2 \\ | \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \end{array}$
 (3) $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH}_2$ and $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$
 (4) $\text{CH}_3 - \text{CH}_2\text{CH}_2 - \text{CH}_2 - \text{CN}$ and $\begin{array}{c} \text{CN} \\ | \\ \text{CH}_3 - \text{CH}_2 - \text{CH} - \text{CH}_3 \end{array}$

63. Minimum number of carbon atoms are required to show chain isomerism in ketone is:

- (1) 4 (2) 5 (3) 3 (4) 6

64. Correct order of elements in decreasing order of electron affinity :-

- (1) $\text{Si} > \text{P} > \text{C} > \text{N}$ (2) $\text{P} > \text{N} > \text{Si} > \text{C}$
 (3) $\text{Si} > \text{C} > \text{P} > \text{N}$ (4) $\text{P} > \text{Si} > \text{C} > \text{N}$

65. Correct order of radius:

- (1) $\text{S}^{-2} > \text{Ca}^{+2} > \text{K}^+ > \text{Cl}^-$
 (2) $\text{S}^{-2} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{+2}$
 (3) $\text{Ca}^{+2} > \text{Cl}^- > \text{S}^{-2} > \text{Cl}^-$
 (4) $\text{Ca}^- > \text{Ca}^{+2} > \text{S}^{-2} > \text{K}^+$

66. Correct order of Ionisation energy is-

- (1) $\text{K}^+ < \text{Cu}^+ < \text{Cu} < \text{K}$ (2) $\text{Cu}^+ < \text{Zn}^{+2}$
 (3) $\text{Zn} < \text{Zn}^+ < \text{Cu} < \text{Cu}^+$ (4) $\text{Tl} > \text{Ga} < \text{Al}$

67. In which pair first element has more Z_{eff} than second atom ?

- (1) Be, B (2) C, N (3) Na, K (4) F, O

68. Which is correct for diborane :-

- (1) Diborane has 12 bonding electrons
 (2) Diborane is dimer of BH_3
 (3) Diborane is a lewis acid
 (4) All of these

69. Which of the following halogen form only one oxyacids (HOX)

- (1) F (2) Br (3) I (4) Cl

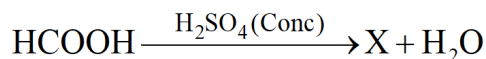
70. Pyrosilicate ion is :

- (1) SiO_2^{2-} (2) SiO_4^{2-} (3) $\text{Si}_2\text{O}_6^{7-}$ (4) $\text{Si}_2\text{O}_7^{6-}$

71. Which of the following is not true for CO ?

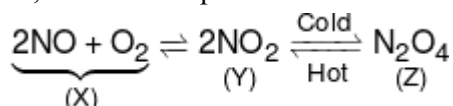
- (1) part of syn gas
 (2) CO is poisonous gas
 (3) CO converts blue litmus paper into red
 (4) CO is a ligand

72. In Given Reaction X is :



- (1) CO_2
- (2) CO
- (3) C_3O_2
- (4) 1 & 2 both

73. X, Y & Z are represented as follow :-



	X	Y	Z
(1)	Colourless	Brown, paramagnetic	Colourless, diamagnetic
(2)	Brown	Colourless, diamagnetic	Brown, paramagnetic
(3)	Colourless	Colourless, paramagnetic	Brown, diamagnetic
(4)	Brown	Brown, paramagnetic	Brown, diamagnetic

74. Bleaching action of O_3 & SO_2 are due to respectively:

- (1) Oxidation, Oxidation
- (2) Reduction, Reduction
- (3) Oxidation, Reduction
- (4) Reduction, Oxidation

75. Select the correct statement regarding $\text{N}(\text{SiH}_3)_3$ compound.

- (i) With respect to Nitrogen it is planar.
 - (ii) Both N and Si has same Hybridisation.
 - (iii) It has $p\pi-d\pi$ Bond.
 - (iv) Bond angle is less than expected due to Back Bonding.
- (1) Only (i) and (ii)
 - (2) Only (i),(iii) and (iv)
 - (3) Only (i) and (iii)
 - (4) Only (iii) & (iv)

76. Formula of Hypo phosphorous acid is -

- (1) H_3PO_2
- (2) H_3PO_3
- (3) H_3PO_4
- (4) H_3PO_5

77. Choose the correct statement

- (1) Aqueous solution of NH_3 turns Blue litmus paper into Red.
- (2) $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \rightarrow$ gas 'x' gas x has Rotten egg smell.
- (3) Reactivity order

$$\text{White Phosphorous} > \text{Red Phosphorous}$$
- (4) Among in N_2O_3 , N_2O_4 , N_2O_5 oxide, N_2O_5 is known as mixed anhydride.

78. Select the Amphoteric oxide -

- (1) CO
- (2) SnO
- (3) N_2O_3
- (4) P_2O_3

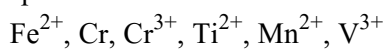
79. Predict the Hydrolysis product of I-Cl molecule -

- (1) HOI , HCl
- (2) HI , HCl
- (3) HI , HOCl
- (4) HOI , HOCl

80. Which of the following compound on heating produce Nitrogen gas.

- (1) $(\text{NH}_4)_2\text{SO}_4$
- (2) $(\text{NH}_4)\text{NO}_2$
- (3) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- (4) Both (2) & (3)

81. Find the total number of species having magnetic moment value of 2.84 B.M. from following species.



- (1) Zero
- (2) 3
- (3) 2
- (4) 4

82. Match the column

	Column I (Characteristic involved in the given process of Column II)		Column II (Process described)
(A)	Energy released	(P)	$S \rightarrow S^{-}$
(B)	Energy absorbed	(Q)	$O^{-} \rightarrow O^{2-}$
(C)	Inert gas configuration is achieved	(R)	$Sr \rightarrow Sr^{2+}$
(D)	Half filled configuration is achieved	(S)	$N^{-} \rightarrow N$
		(T)	$Ge \rightarrow Ge^{-}$

- (1) (A→S,T), (B→Q,R), (C→Q,R), (D→T)
 (2) (A→P,S,T), (B→Q,R), (C→Q,R), (D→ST)
 (3) (C→T), (B→R), (C→R), (D→ST)
 (4) (C→P,S,T), (B→QR), (C→ST), (D→QR)

83. Consider the $M(OH)_3$ formed by all the group 13 element. The correct sequence of acidic strength of hydroxides $[M(OH)_3]$ is -

- (1) $B(OH)_3 < Al(OH)_3 > Ga(OH)_3 > In(OH)_3 > Tl(OH)_3$
 (2) $B(OH)_3 > Tl(OH)_3 > Al(OH)_3 > In(OH)_3 > Ga(OH)_3$
 (3) $Al(OH)_3 > Ga(OH)_3 > B(OH)_3 > In(OH)_3 > Tl(OH)_3$
 (4) $B(OH)_3 > Al(OH)_3 > Ga(OH)_3 > In(OH)_3 > Tl(OH)_3$

84. Increasing order of basic character of NO_2 , K_2O and ZnO is

- (1) $NO_2 < ZnO < K_2O$
 (2) $K_2O < ZnO < NO_2$
 (3) $ZnO < K_2O < NO_2$
 (4) $ZnO < NO_2 < K_2O$

85. Which of the following should have longest bond ?

- (1) S-H (2) O-H (3) N-H (4) P-H

SECTION-B (CHEMISTRY)

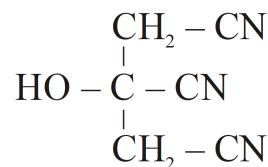
86. The IUPAC name of isobutyl alcohol :

- (1) Butan-1-ol (2) Butan-2-ol
 (3) 2-methyl propan-2-ol (4) 2-methyl propan-1-ol

87. Correct IUPAC name of Resorcinol?

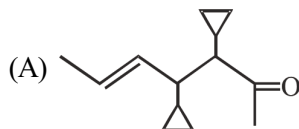
- (1) 1,2-dihydroxy benzene (2) 1,3-dihydroxy benzene
 (3) Benzene-1, 2-diol (4) Benzene-1,3-diol

88. IUPAC name of given compound ?

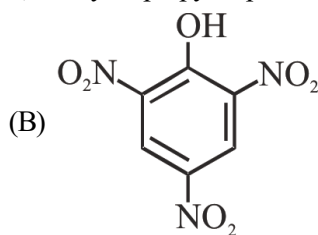


- (1) 3-cyano-2-hydroxy pentanedinitrile
 (2) 2-hydroxy propane - 1, 2, 3-tricarbonitrile
 (3) Propane - 1, 2, 3-tricarbonitrile
 (4) Propane - 1, 2, 3-tricarbonitrile-2-ol

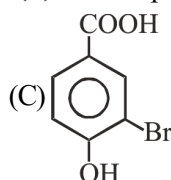
89. Which of the following is/are correct :



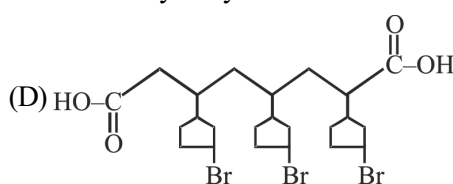
3,4-Dicyclo propyl hept-5-en-2-one



2,4,5-trinitro phenol



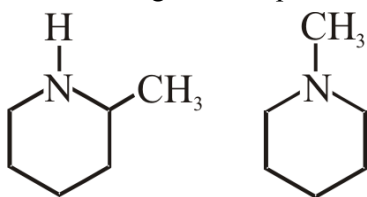
3-Bromo-4-Hydroxy Benzoic acid



2,4,6-tris(3-Bromo cyclopentyl) octane-1, 8-dioic acid

- (1) ABCD (2) ACD (3) CD (4) AD

90. The relation between given compound is :-



- (1) Chain isomerism (2) Position isomerism
(3) Functional isomerism (4) Metamerism

91. If an atom has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$, it will be placed in :

- (1) Second group (2) Third group
(3) Fifth group (4) Sixth group

92. Correct order of acidic properties ?

- (1) $SO_3 < SO_2$ (2) $Sb_2O_5 > Sb_2O_3$
(3) $N_2O_5 < N_2O_3$ (4) $HNO_3 < HNO_2$

93. Hydrolysis of Magnesium carbide will give, Hydrocarbon :-

- (1) C_3H_6 (2) C_3H_4 (3) C_2H_2 (4) CH_4

94. Which compound is not a monobasic acid ?

- (1) H_3PO_2 (2) H_3BO_3 (3) HCl (4) H_3PO_3

95. Molecule which has maximum Bond angle -

- (1) OH_2 (2) $N(SiH_3)_3$ (3) OF_2 (4) NH_3

96. Which of following has crown structure -

- (1) White Phosphorous (2) Rhombic sulphur
(3) Graphite (4) fullerene

97. Match the column

	Column I (Orders)		Column II (Sequences)
(A)	Increasing order of ionic mobility in water	(P)	Si, Mg, Al, Na
(B)	Decreasing order of size	(Q)	Li^+ , Na^+ , K^+ , Rb^+ , Cs^+
(C)	Decreasing order of ionisation energy	(R)	O , O^+ , O^{2-} , O^-
		(S)	Se, S, O, F

- (1) $(A \rightarrow Q), (B \rightarrow S), (C \rightarrow P, Q)$
(2) $(A \rightarrow S), (B \rightarrow Q), (C \rightarrow P, Q)$
(3) $(A \rightarrow Q), (B \rightarrow S), (C \rightarrow Q)$
(4) $(A \rightarrow Q), (B \rightarrow P, Q), (C \rightarrow S)$

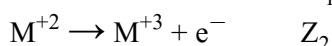
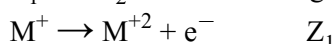
98. **Statement-I** : Cs and F combines violently to form CsF.

Statement-II : Cs is most electropositive and F is most electronegative.

These Questions you are required to choose any one of the following four response

- (1) Both statement-I & statement-II are True & the statement-II is a correct explanation of the statement-I.
(2) Both statement-I & statement-II are True but statement-II is not a correct explanation of the statement-I.
(3) Statement-I is True but the statement-II is False.
(4) Statement-I is False but the statement-II is True.

99. Which of the following should be correct for Z_1 and Z_2 in the following two processes



- (1) $\frac{1}{2}Z_1 = Z_2$
(2) $Z_1 = Z_2$
(3) $Z_1 = \frac{1}{2}Z_2$
(4) $Z_1 < Z_2$

100. The nuclei of elements X, Y and Z have same number of protons, but different numbers of neutrons. According to Mendeleev's periodic table, the elements X, Y and Z

- (1) belong to same group and same period
(2) belong to different groups and different period
(3) belong to same group and different periods
(4) are isotopes, which do not have different positions.

Topic : HUMAN GENOME PROJECT, PRINCIPLES OF INHERITANCE AND VARIATIONS, MENDELIAN GENETICS, ECOLOGY (POLLUTION, BIODIVERSITY), DEMOGRAPHY, PHOTO SYNTHESIS, PLANT MORPHOLOGY [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM

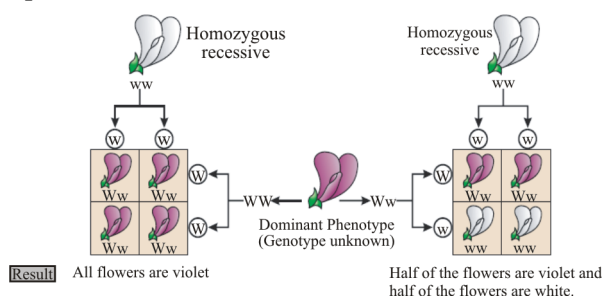
TG: @Chalnaayaaar

SECTION-A (BIOLOGY-I)

101. (A) A true breeding line is one that have undergone continuous self pollination
(B) Show stable trait inheritance and expression for several generation
(C) Mendel selected 7 true breeding pea plant varieties as pair which were dissimilar except for one character with contrasting traits
Regarding to pure lines select the correct statement

- (1) A, B and C (2) A and B
(3) A and C (4) B and C

102. Observe the given diagram & select the correct option :-



- (1) It is the diagrammatic representation of a out cross.
(2) It is the comparison of two monohybrid crosses, used to find out the phenotype of organisms.
(3) It is the diagrammatic representation of reciprocal cross to find out the genotype of organism.
(4) It is the diagrammatic representation of test cross to find out the genotype of organisms.
103. Which of the following statement is/are correct regarding law of segregation ?
- (1) Alleles separate from each other during gametogenesis
(2) The segregation of factors is due to the segregation of chromosomes during meiosis
(3) Law of segregation is also called as law of purity of gametes
(4) All of the above

104. Which of the following crosses will give tall and dwarf pea plants in same proportions ?

- (1) $TT \times tt$ (2) $Tt \times tt$
(3) $TT \times Tt$ (4) $tt \times tt$

105. Heterozygous tall plants of garden pea were self pollinated. In the progeny 520 plants were produced. How many of these plants were homozygous?

- (1) 390 (2) 260 (3) 130 (4) 412

106. In the Punnett square given below, the genotype of female parent will be ?

		Male gametes			
Female gametes	?	AB	Ab	aB	ab
		AABb	AAbb	AaBb	Aabb

- (1) AaBb (2) AAbb
(3) aaBB (4) Aabb

107. In F_2 generation of a dihybrid cross, what is the probability of offsprings having parental phenotype ?

- (1) $9/16$ (2) $2/16$ (3) $1/16$ (4) $10/16$

108. When a cross is made between tall plant with yellow seed ($TtYy$) and tall plant with green seed ($Ttyy$), what percentage of offspring could be expected to be tall and green.

- (1) 25% (2) 12.5% (3) 37.5% (4) 50%

109. Read the following statements with respect to back cross ?

- (A) It is the cross of F_1 individual with either one of the two homozygous parents
(B) Phenotypic & genotypic ratio is always same
(C) It includes both test cross and out cross
(D) It was performed by Mendel in his breeding experiment

How many statements are correct ?

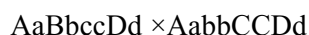
- (1) One (2) Two (3) Three (4) Four

110. Read the following statements regarding mendelian inheritance and choose the correct option :

- A. Mendel's experiments had small sample size which gave greater credibility to the data he collected.
- B. Segregation of alleles occur during anaphase-I.
- C. In a dissimilar pair of factors, one member of the pair dominates over the other.
- D. A recessive parental trait is expressed only in its heterozygous condition.
- E. Two alleles of a gene are located at same locus on homologous chromosomes.

- (1) B alone is correct
- (2) B, C and E are correct
- (3) A and D are correct
- (4) A, C and E are correct

111. In the cross given below :



What percentage of progeny will have genotype aaBbCcDD?

- (1) 6.2 (2) 3.1 (3) 12.4 (4) 25

112. In a plant species, flower colour yellow dominant over white and fruit shape round is dominant over elongated. Crossing was performed between two pure lines one having yellow flower and round fruit and another with white flower and elongated fruit. About 20 plants survived in F_1 progeny. Plant of F_1 were allowed to self fertilize and about 960 plants survived in F_2 . If the traits follow Mendelian inheritance the number of plant that would have yellow flower and round fruit in F_1 and F_2 are respectively

- (1) 20, 960 (2) 20, 540
- (3) 10, 180 (4) 10, 60

113. A cross between plants heterozygous for two different genes ($AaBb \times AaBb$) each assorting independently. What fraction of progeny will show dominant phenotype for atleast one character ?

- (1) $\frac{6}{16}$ (2) $\frac{9}{16}$ (3) $\frac{7}{16}$ (4) $\frac{15}{16}$

114. Match the column-I (Type of offsprings) with the column-II (Their probability) in F_2 generation of a Mendelian dihybrid cross.

Column-I		Column-II	
A	Homozygous for one and Heterozygous for other character.	i	6/16
B	Recombinant phenotype	ii	2/16
C	F_1 phenotype	iii	8/16
D	Parental Genotypes	iv	9/16

- (1) A-iv, B-ii, C-i, D-iii
- (2) A-iii, B-i, C-ii, D-iv
- (3) A-i, B-iii, C-iv, D-ii
- (4) A-iii, B-i, C-iv, D-ii

115. Which one of the following is not correct regarding Mendelian dihybrid cross ?

- (1) The F_2 phenotypic ratio is 9 : 3 : 3 : 1
- (2) Law of independent assortment can be explained on the basis of the cross
- (3) F_2 generation is obtained from the cross between F_1 plants with one of the two parental plants
- (4) The gametes produced by F_1 plants are of 4 types in 1 : 1 : 1 : 1 ratio

116. A male human is heterozygous for autosomal genes A and B and is also hemizygous for haemophilic gene h. What fraction of his sperm will have abh ?

- (1) $\frac{1}{32}$ (2) $\frac{1}{4}$ (3) $\frac{1}{16}$ (4) $\frac{1}{8}$

117. In corn, the trait for tall plant (T) is dominant to the trait for dwarf plant (t) and the trait for coloured kernels (C) is dominant to the trait for white kernels (c). In a particular cross of corn plant, the probability of an offspring being tall is 0.5 and probability of a kernel being coloured is 0.75. Which of the following most probably represents the parental genotypes?

- (1) $TtCc \times TtCc$ (2) $TtCc \times ttCc$
- (3) $TtCc \times ttcc$ (4) $TTCc \times ttCc$

118. Which of the following structure would be absent in palmately compound leaf?

- (1) Leaflet (2) Petiole
(3) Rachis (4) Vein

119. Which of the following is **incorrect** :-

- (1) Reticulate venation is characteristic of most dicotyledonous plants
(2) Parallel venation is characteristic of most monocotyledons
(3) In neem pinnately compound leaf is found
(4) In silk cotton pinnately compound leaf is found

120. Which type of placentation is found in solanaceae and liliaceae?

- (1) Marginal (2) Axile
(3) Parietal (4) Basal

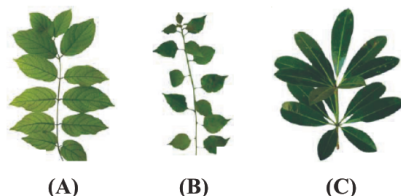
121. (9) + 1 arrangement of stamens with superior ovary is the characteristic feature of family :-

- (1) Liliaceae
(2) Brassicaceae
(3) Solanaceae
(4) Fabaceae

122. Diadelphous condition of stamens is found in :-

- (1) China rose (2) Brinjal
(3) *Brassica* (4) Pea

123.



Identify the examples of given phyllotaxy, respectively :-

- (1) Chinrose, *Calotropis*, *Alstonia*.
(2) *Alstonia*, Chinrose, *Calotropis*.
(3) *Calotropis*, Chinrose, *Alstonia*.
(4) Mustard, Guava, *Alstonia*.

124. Which of the following match is incorrect ?

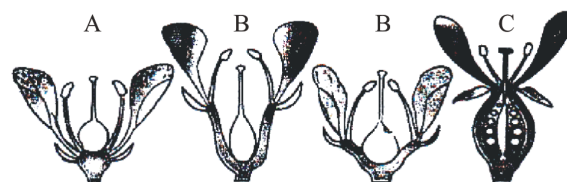
Column-I		Column-II	
(1)		(a)	Valvate
(2)		(b)	Twisted
(3)		(c)	Quincuncial
(4)		(d)	Vexillary

125. Match the following figures with I, II and III.

I-Hypogynous flower

II-Perigynous flower

III-Epigynous flower



- (1) A-I, B-II, C-III
(2) A-I, B-III, C-II
(3) A-III, B-II, C-I
(4) A-III, B-I, C-II

126. Which of the following is adopted as substitute for open burning dumps?

- (1) Incineration
(2) Discarding waste in oceans
(3) Sanitary Land fills
(4) Composting

127. Which of the following pairs of gases is mainly responsible for global warming ?
- Ozone and Ammonia
 - Oxygen and nitrogen
 - Nitrogen and sulphur dioxide
 - CO₂ and methane
128. As the exhaust passes through the catalytic converter, carbon monoxide and nitric oxide respectively are converted to
- Carbon dioxide and water
 - Carbon dioxide and nitrogen gas
 - Water and nitrogen gas
 - Carbon dioxide and nitrous oxide
129. In which among the given organisms of aquatic food chain, biomagnification of heavy metals would be least in:-
- Fish eating birds
 - Zooplankton
 - Small fish
 - Large fish
130. How many statements are correct for Noise pollution :-
- Noise is undesired high level of sound
 - Noise causes sleeplessness and alter breathing pattern
 - Noise is measured in dobson units.
 - Schools and Hospitals are horn-free zone
- a, b, d
 - b, c, d
 - a, c, d
 - a, b, c & d
131. Which of the following statements are correct ?
- Eutrophication is natural aging of a lake by nutrient enrichment of its water
 - After CO₂, methane (CH₄) is major cause of green house effect
 - Ozone is secondary pollutant in troposphere
 - Thickness of ozone is measured in dobson unit.
- a, b only
 - a, b, c only
 - a, b, c, d
 - b, c, d only

132. Kyoto protocol is to :-
- mitigate the climatic change
 - check depletion of ozone layer
 - check soil erosion
 - preserve water resources
133. Which of the following conditions show the polluted environment ?
- pH of rain water is lower than 5.6
 - Amount of CO₂ in the air is 0.03 %
 - BOD high
 - Accelerated Eutrophication
- abc
 - ab only
 - cd only
 - acd
134. Which of the following is a result of improper resource utilisation and maintenance ?
- Soil erosion
 - Desertification
 - Water logging and soil salinity
 - All of the above
135. Match the column given below :-

	Column A		Column B
(a)	Restoring forest that once existed	(i)	1989
(b)	National forest policy	(ii)	Reforestation
(c)	Sacred grove	(iii)	1988
(d)	Montreal protocol effective in	(iv)	Aravalli hill

- a-ii, b-i, c-iv, d-iii
- a-iv, b-i, c-ii, d-iii
- a-iv, b-iii, c-ii, d-i
- a-ii, b-iii, c-iv, d-i

SECTION-B (BIOLOGY-I)

TG: @Chalnaayaaar

136. Find an incorrect statement with respect to findings of HGP :

- (1) Dystrophin gene is largest gene of human genome
- (2) Chromosome - Y has minimum 231 genes
- (3) SNPs were identified at about 2.4 million locations
- (4) H.G.P was closely associated with development of Bioinformatics

137. In human

- I. non-coding DNA is abundant.
- II. less than 2% of genome codes for protein.
- III. the function of more than 50% genes are unknown
- IV. total number of genes is 30000.

Correct option are :-

- (1) I, II, III and IV
- (2) I and III
- (3) I, II and IV
- (4) I, II and III

138. In HGP, the blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequences, is known as

- (1) Expressed Sequence Tags
- (2) ELSI
- (3) Single nucleotide polymorphism
- (4) Sequence Annotation

139. The segregation of allele is a random process and so there is a ____ chance of a gamete containing either allele.

- (1) 50%
- (2) 25%
- (3) 75%
- (4) 100%

140. Among the seven pair of contrasting trait in pea plant as studied by Mendel, the number of character related to flower, pod and seed respectively were :-

- (1) 2, 2, 2
- (2) 2, 2, 1
- (3) 1, 2, 2
- (4) 1, 1, 2

141. The punnett square shown below represents the pattern of inheritance in dihybrid cross :-

	RY	Ry	rY	ry
RY	F	J	N	R
Ry	G	K	O	S
rY	H	L	P	T
ry	I	M	Q	U

Choose the correct symbol having similar genotype

- (1) F, K
- (2) G, H
- (3) L, O
- (4) P, U

142. Match the columns and choose correct option:-

Cross		Types of gametes		Types of Phenotype		Types of genotype		Total zygotic combinations	
A	Monohybrid	a	4	i	4	p	9	x	64
B	Dihybrid	b	8	ii	2	q	3	y	4
C	Trihybrid	c	2	iii	8	r	27	z	16

- (1) A-a-i-p-x, B-b-ii-q-y, C-c-iii-r-z
- (2) A-c-ii-p-y, B-a-iii-r-x, C-b-i-q-z
- (3) A-b-i-q-y, B-c-ii-p-z, C-a-iii-r-x
- (4) A-c-ii-q-y, B-a-i-p-z, C-b-iii-r-x

143. Study the sentences carefully and give the answer.
 (A) When a flower can be divided into two equal radial halves through any plane, it is zygomorphic.
 (B) Mustard, datura and chilli are examples of actinomorphic flowers
 (C) Pea, gulmohur, bean, *Cassia* are examples of zygomorphic flowers.
 (D) If a fruit is formed without fertilisation of the ovary, it is called geocarpic fruit.

- (1) Both A and B are correct
 (2) Both C and D are correct
 (3) Both A and D are incorrect
 (4) Both B and D incorrect

144. Fleshy thalamus is edible part of fruit in :-

- (1) Peach, Plum, Rose
 (2) Apple, Pear, Strawberry
 (3) Litchi, Apple, Guava
 (4) Fig, pineapple, Mulberry

145. In *Opuntia*, the function of photosynthesis is carried out by :-

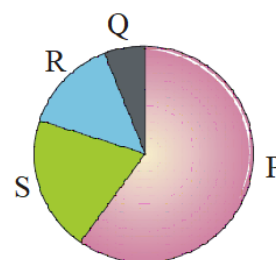
- (1) Cladode (2) Phylloclade
 (3) Phyllode (4) Bulb

146. Match the column-I with Column-II :-

I		II	
i	Oxide of sulphur	a	Global warming
ii	Nitrogen oxide	b	Plumbism
iii	CO ₂	c	Blue baby syndrome
iv	Nitrate in drinking water	d	London smog
v	Pb	e	Photochemical smog

- (1) i → b; ii → a; iii → e; iv → c; v → b
 (2) i → d; ii → e; iii → a; iv → c; v → b
 (3) i → d; ii → b; iii → a; iv → c; v → e
 (4) i → c; ii → b; iii → a; iv → d; v → e

- 147.



Above graph denotes the relative contribution of various green house gases to global warming. Identify P,Q,R,S.

- (1) P = CO₂-60%; Q = CFC-14%; R = N₂O-6%; S = CH₄-20%
 (2) P = CO₂-60%; Q = N₂O-6%; R = CFC-14%; S = CH₄-20%
 (3) P = CO₂-60%; Q = CH₄- 6%; R = CFC = 20%; S = N₂O-14%
 (4) P = CO₂-60%; Q = CFC-14%; R = N₂O-6%; S = CH₄-30%

148. The UN conference of parties on climate change in the year 2012 was held at :-

- (1) Warsaw (2) Durban (3) Lima (4) Doha

149. Choose the correct matches for domestic sewage :-

	Suspended solids-I	Colloidal Material-II	Dissolved Material-III
A	Sodium & Calcium	Silt & clay	Fecal matter bacteria
B	Silt & Clay	Fecal matter	Phosphate, Nitrate
C	Sand	Cloth & fibers paper	Sodium & calcium
D	Phosphate calcium	Cloth fiber & fecal matter	Sodium Ammonia

- (1) Only B is correct (2) B & C are correct
 (3) A and D are correct (4) C & D are correct

150. Wet lands occupy about _____ % of world's land surface.

- (1) 16% (2) 14% (3) 6% (4) 19.4%

Topic : HUMAN GENOME PROJECT, PRINCIPLES OF INHERITANCE AND VARIATIONS, MENDELIAN GENETICS, ECOLOGY (POLLUTION, BIODIVERSITY), DEMOGRAPHY, PHOTO SYNTHESIS, PLANT MORPHOLOGY [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM

TG: @Chalnaayaaar

SECTION-A (BIOLOGY-II)

151. Match the column-I with column-II

I-(Activity)		II- Effects	
i	Releasing gases to the atmosphere after burning waste material, containing sulphur	a	Water pollution and eutrophication
ii	Using chemical pesticides	b	Photochemical smog damage to plant life, corrosion in building material, induce breathing problems
iii	Using synthetic detergents for washing cloths	c	Damaging ozone layer
iv	Releasing gases produced by automobiles and factories in the atmosphere	d	May cause biomagnification in human
v	Using CFC compounds	e	Classical Smog acid rain, breathing problems, damage to buildings

- (1) (i) d, (ii) e, (iii) b, (iv) a, (v) c
- (2) (i) e, (ii) d, (iii) a, (iv) b, (v) c
- (3) (i) a, (ii) b, (iii) c, (iv) d, (v) e
- (4) (i) b, (ii) c, (iii) a, (iv) b, (v) e

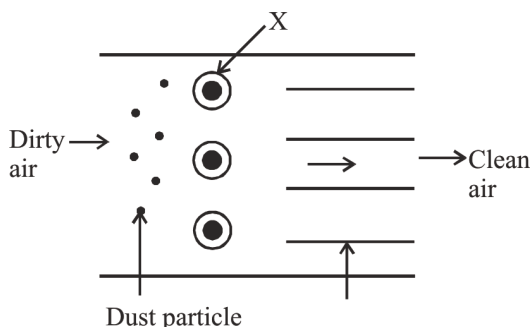
152. Correctly match the following column and choose the correct option

Column -I		Column-II	
i	Environment Protection Act	A.	1974
ii	Air Prevention & Control of Pollution Act	B.	1987
iii	Water Act	C.	1986
iv	Amendment of Air Act to include noise pollution	D.	1981

The correct matches is

- (1) i-C, ii-D, iii-A, iv-B
- (2) i-A, ii-C, iii-B, iv-D
- (3) i-D, ii-A, iii-B, iv-C
- (4) i-C, ii-D, iii-B, iv-A

153.



In following diagram of electrostatic precipitator 'X' is

- (1) Corona discharge
- (2) Positively charged wire
- (3) Lime spray
- (4) Collection plate grounded

154. COP-21 is also known for :-

- (1) The Earth summit
- (2) The Paris agreement
- (3) The Montreal protocol
- (4) The Kyoto protocol

155. Match the column I and II :-

Column-I		Column-II	
(a)	Global animal diversity (%)	(i)	1200
(a)	Global plant diversity (%)	(ii)	8.1
(c)	Species diversity in India (%)	(iii)	22
(d)	Bird species in India	(iv)	70

- (1) (a)-iii, (b)-iv, (c)-i, (d)-ii
 (2) (a)-iv, (b)-ii, (c)-iii, (d)-i
 (3) (a)-iv, (b)-iii, (c)-ii, (d)-i
 (4) (a)-iii, (b)-ii, (c)-i, (d)-iv

156. Which of the following statement(s) is/are incorrect ?

- (A) Large proportion of the species waiting to be discovered are in temperate areas.
 (B) Conventional taxonomic methods are well suitable for identification of microbial species.
 (C) Number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals
 (D) India's share of global species diversity is 8.1 percent

- (1) B only (2) A, B and C
 (3) A and B (4) A and C

157. (i) Sacred groves

(ii) Hot spots

(iii) National park

(iv) Wild life safari park

(v) Seed bank

(vi) Biosphere reserve

(a) i, ii, iii, vi are the example of insitu conservation

(b) iv, v are the example of Ex-situ conservation

(c) ii, iv, v are the example of Ex-situ conservation

(d) i, ii, iii, iv, vi are the example of insitu conservation

- (1) a, b are correct (2) a, c are correct
 (3) c, d are correct (4) a, d are correct

158. Usually in any island ecosystem, introduction of exotic species causes :-

- (A) Disappearance of native species
 (B) Increase negative interactions
 (C) Increase in biodiversity
 (D) Increase in productivity

- (1) A, B, C and D
 (2) B, C and D
 (3) A, C and D
 (4) A and B

159. National forest policy (1988) of india has recommended % of forest cover for the plains and % for the hills :-

- (1) 33; 19.4 (2) 33; 67
 (3) 19.4; 33 (4) 33, 100

160. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
(i)	Jhum cultivation	North-eastern states of India	Slash and Burn agriculture
(ii)	Chipko movement	Garhwal Himalaya	1974
(iii)	Joint forest management	1980 s	Forest conservation with local communities

- (1) (i) only (2) (i) and (ii)
 (3) (i), (ii) and (iii) (4) (i) and (iii)

161. Which of the following is most important factor determining population growth if a habitat is just being colonised :-

- (1) Emigration (2) Birth rate
 (3) Death rate (4) Immigration

162. Match the following

(a)	Breeding once in life	(i)	Mammals
(b)	Breeding several time in life	(ii)	Oysters
(c)	Large number of small size offspring	(iii)	most of birds
(d)	Small number of large size offspring	(iv)	Pacific fish salmon

- (1) a(iv), b(ii), c(iii), d(i)
 (2) a(iv), b(iii), c(ii), d(i)
 (3) a(iii), b(iv), c(ii), d(i)
 (4) a(ii), b(iii), c(iv), d(i)

163. Which among the following statement(s) is/are true ?

- (A) According to rivet popper hypothesis, rivets of air plane are key stone species that drive major ecosystem functions.
 (B) According to David Tilman, Communities with more species show less year to year variation in total biomass.
 (C) According to David Tilman, increased diversity contributed to higher productivity.

- (1) A only (2) B & C
 (3) A, B, C (4) C only

164. Study the following statements regarding chl. a molecule :

- (i) Molecular formula of chl 'a' is $C_{55}H_{72}O_5N_4Mg$.
 (ii) It is the chief photosynthetic pigment.
 (iii) It absorbs more blue wavelength of light than the red wavelength.
 (iv) It is soluble in water as well as petroleum ether.

Which of the above statements is/are not correct ?

- (1) (i) and (iii) (2) (iii) and (iv)
 (3) (iii) only (4) (iv) only

165. Which of the following is a perennial monocarpic plant ?

- (1) *Strobilanthus kunthiana*
 (2) Wheat
 (3) Maize
 (4) All of the above

166. Long, ribbon like pollen grains are seen in some:-

- (1) Bird pollinated plants
 (2) Bat pollinated plant
 (3) Gymnosperms
 (4) Aquatic plants

167. Microsporogenesis occurs :-

- (1) On margins of leaves.
 (2) Inside the ovule.
 (3) Inside the anther.
 (4) In accessory floral organs.

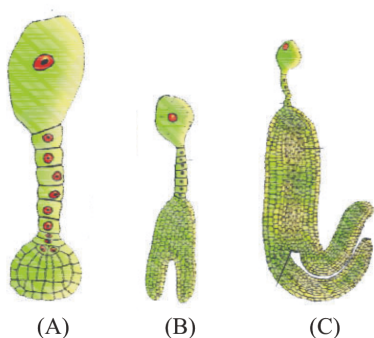
168. How many cells of nucellus of each ovule in *Capsella* are responsible to form megaspore mother cell ?

- (1) Many
 (2) Four
 (3) Two
 (4) One

169. Which of following event can never take place after the degeneration of generative cell in pollen grain:-

- (1) Entry of pollen tube inside the embryo sac
 (2) Fertilization
 (3) Only triple fusion but not syngamy
 (4) Neither entry of pollen-tube inside the embryo sac nor fertilization.

170. Identify the correct sequence of stage of embryo development :



- (1) A—Quadrant, B—Heart shape, C—Torpedo
 (2) A—Octate, B—Heart, C—Torpedo
 (3) A—Octate, B—Torpedo, C—Mature
 (4) A—Globular, B—Heart shape, C—Mature
171. (a) Pollen tube shows chemotropic movement in ovary.
 (b) In porogamy, pollen tube enters into the ovule through the micropyle.
 (c) Male gametes of angiosperms are non motile.
 (d) Double fertilisation and triple fusion is the specific or universal characteristics of Angiosperms. Choose the correct statements.
- (1) a,b,c only
 (2) a,c & d only
 (3) a,b,c & d
 (4) b,c,d only
172. Mango, Peepal, *Commelina*, *Oxalis*, *Viola*
 How many plants from above list show assured seed set even in absence of pollinators :-
- (1) Four (2) Three (3) Two (4) One
173. Pollen grain exine has prominent apertures called __A__ where sporopollenin is __B__.
- (1) A-Pollen tube, B-Ornamented
 (2) A-Germ pore, B-Absent
 (3) A-Germ pore, B-Present
 (4) A-Intine, B-Absent

174. Which of the following is an out breeding device ?
- (1) Cleistogamy
 (2) Homogamy
 (3) Self incompatibility
 (4) Bud pollination
175. When pollen grain mature it contains two cells, they are :-
- (1) Two motile male gametes
 (2) Synergid cells
 (3) Egg cell and male gamete
 (4) Vegetative cell and generative cell
176. **Statement I** : Flowers are the seat of sexual reproduction in angiosperms.
Statement II : All seeded plants shows flowering atleast once their life span.
- (1) Both statement I and statement II are correct
 (2) Statement I is correct but statement II is incorrect
 (3) Statement I is incorrect but statement II is correct
 (4) Both statement I and statement II are not correct
177. Which of the following serves as landing platform for pollen grain ?
- (1) Anther
 (2) Ovary
 (3) Ovule
 (4) Stigma
178. Pollen grains maintain viability for months in :-
- (1) Rosaceae
 (2) Solanaceae
 (3) Leguminosae
 (4) All of the above

179. If there are 24 chromosomes are present in endosperm of an angiosperm, then find out the number of chromosomes in antipodal cell of embryo sac :-

- (1) 12 (2) 6 (3) 8 (4) 16

180. When tetraploid male plant is crossed with a tetraploid female plant, the ploidy of endosperm and embryo will be respectively :-

- (1) Tetraploid and hexaploid
(2) Pentaploid, and hexaploid
(3) Hexaploid and tetraploid
(4) Tetraploid and pentaploid

181. When egg cell develops into the embryo without fertilisation, the process is known as :-

- (1) Parthenocarpy (2) Parthenogenesis
(3) Apogamy (4) Polyembryony

182. In angiosperms correct sequence of events in formation of female gametophyte in the ovule is :

- (I) Degeneration of 3 megaspores.
(II) Meiotic division in megaspore mother cell.
(III) 3 successive free nuclear divisions in functional megaspore.
(IV) Formation of wall resulting in seven celled embryo sac.

Choose the correct answer from the options given below :

- (1) (II), (I), (IV), (III)
(2) (IV), (III), (II), (I)
(3) (II), (I), (III), (IV)
(4) (I), (II), (III), (IV)

183. Pollination by wind is :

- (1) Limited to about 30 genera of angiosperms
(2) Quite rare in grasses
(3) Most common amongst abiotic pollinators
(4) Found in water lily and *Vallisneria*

184. Identify the incorrect statement :-

- (1) The number of ovules in an ovary may be one to many.
(2) The ovule is attached to placenta by means of a stalk called hilum.
(3) An ovule generally has a single embryo sac formed from a megaspore.
(4) Chalaza represents the basal part of ovule.

185. Transfer of pollen grains from anther to stigma of a different plant is known as :

- (1) Xenogamy
(2) Autogamy
(3) Cleistogamy
(4) Geitonogamy

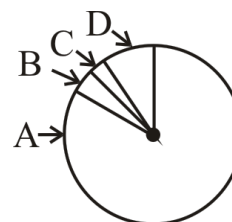
SECTION-B (BIOLOGY-II)

186. Which is/are Incorrect ?

- | | | |
|----------------------------------|---|----------------------------|
| (a) Number of birds species | > | Number of Reptiles species |
| (b) Number of Amphibians species | > | Number of Mammals |
| (c) Number of Mammals species | > | Number of Amphibians |
| (d) Number of Fishes species | > | Number of Insects species |

- (1) only a (2) a, b
(3) c, d (4) a, b, c, d

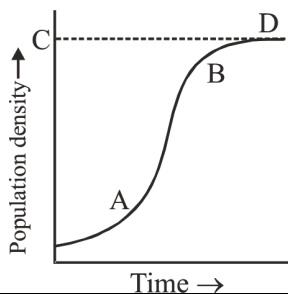
187. In given below pie diagram Identify 'C'



Above diagram represent global biodiversity

- (1) Insects (2) Crustaceans
(3) Molluscs (4) Birds

188. Which of the following is a correct match ?



	Column-I	Column-II
(i)	Carrying capacity	A
(ii)	Deceleration	B
(iii)	Asymptote	C
(iv)	Acceleration	D

Options :-

- (1) (i)-B, (ii)-D, (iii)-C, (iv)-A
- (2) (i)-B, (ii)-C, (iii)-D, (iv)-A
- (3) (i)-D, (ii)-C, (iii)-A, (iv)-B
- (4) (i)-C, (ii)-B, (iii)-D, (iv)-A

189. Identify the correct match from columns I, II and III :-

	Column-I		Column-II		Column-III
1	Chlorophyll 'a'	a	Absorption peak at 680 nm	i	Prevents Photo-oxidation
2	Photosystem-I	b	Accessory pigments	ii	Only grana Lamellae
3	Photosystem-II	c	Chief pigment	iii	Blue green in colour
4	Carotenoids	d	Absorption peak at 700 nm	iv	Present in both grana and stroma lamellae

- (1) 1 - c - iii, 2 - d - iv, 3 - a - ii, 4 - b - i
- (2) 1 - b - i, 2 - a - ii, 3 - c - iii, 4 - d - iv
- (3) 1 - b - iv, 2 - c - iii, 3 - d - ii, 4 - a - i
- (4) 1 - b - i, 2 - a - ii, 3 - d - iii, 4 - c - iv

190. **Assertion:** Clear division of labour is present within chloroplast.

Reason: Grana is responsible for trapping the light energy for synthesis of ATP and NADPH₂ and in stroma enzymatic reactions incorporate CO₂ for synthesis of sugar.

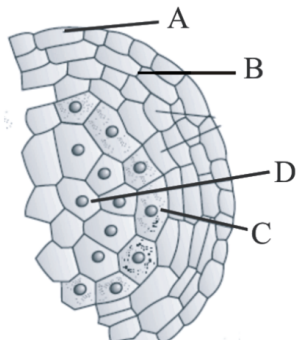
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

191. Match the column and choose the correct option-

	Column-I		Column-II
(i)	Essentiality of CO ₂ is photosynthesis	(a)	C.V. Neil
(ii)	Redox Nature of photosynthesis	(b)	J. Priestley
(iii)	Role of air in growth of plant	(c)	Moll's half leaf experiment.
(iv)	Discoverer of light reaction.	(d)	Robert Hill

- (1) i - c, ii - a, iii - b, iv - d
- (2) i - c, ii - b, iii - a, iv - d
- (3) i - d, ii - a, iii - b, iv - c
- (4) i - c, ii - d, iii - a, iv - b

192. Depicted picture is an enlarged view of microsporangium in which labelled structure A, B, C and D are respectively –



- (1) Tapetum, epidermis, endothecium, middle layer
 (2) Epidermis, endothecium, tapetum, microspore mother cell
 (3) Endothecium, middle layer, tapetum, epidermis
 (4) Endothecium, middle layer, tapetum, microspore mother cell
193. For the formation of female gametophyte, the megaspore mother cell undergoes :-
- (1) Two meiotic and two mitotic divisions
 (2) One meiotic and three mitotic divisions
 (3) Two meiotic divisions
 (4) One meiotic and two mitotic divisions

Column I		Column II	
a	Sporopollenin	i	Formed by generative cell
b	Intine	ii	Formed by vegetative cell
c	Germ pore	iii	Forms exine
d	Pollen tube	iv	Thin, elastic and delicate
		v	Absence of intine
		vi	Absence of sporopollenin

Match the column and choose correct option :-

- (1) a–iii, b–ii, c–iv, d–i
 (2) a–iii, b–iv, c–vi, d–ii
 (3) a–iii, b–iv, c–i, d–ii
 (4) a–iii, b–iv, c–v, d–vi

195. Animal visitors are required for pollination in :-

- (A) *Zostera* (B) Bamboo
 (C) Cucumber (D) Maize
 (E) Pea (F) Ficus
 (G) *Amorphophallus*

- (1) A, B and C
 (2) B, C and G
 (3) C, D and E
 (4) C, F and G

196. Match the column–I with Column–II:-

	Column-I		Column-II
I.	Tapetum	(a)	Ephemeral
II.	Epidermis	(b)	Hygroscopic
III.	Middle layer	(c)	Multinucleate
IV.	Endothecium	(d)	Protective layer

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	I	II	III	IV
(1)	a	c	d	b
(2)	c	d	a	b
(3)	a	b	c	d
(4)	c	d	b	a

197. If the hybrid seeds are made into apomictic seeds what is the advantage ?

- (1) The plant in the progeny will segregate and do not maintain hybrid characters.
 (2) There is no segregation of characters in the hybrid progeny.
 (3) The farmer can not use apomictic seed to raise new crops year after year.
 (4) Farmer have to buy hybrid seed every year.

198. How many plants in the list given below have multicarpellary syncarpous gynoecium ?

Rose, Lotus, *Michelia*, *Papaver*.

(1) Three (2) Two (3) Four (4) One

199. Generally embryo and endosperm of angiosperms respectively are :-

(1) Diploid and diploid

(2) Triploid and diploid

(3) Diploid and triploid

(4) Haploid and triploid

200. **Assertion** : In some species floral rewards are in providing safe places to lay eggs.

Reason : To sustain animal visits, the flowers have to provide rewards to the animals.

(1) Both assertion and reason are true but reason is not the correct explanation of assertion

(2) Assertion is true but reason is false

(3) Assertion is false but reason is true

(4) Both assertion and reason are true and reason is the correct explanation of assertion

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